

Basic Pharmacology of the Antiarrhythmic Agents

Introduction

- ❖ Impulse conduction in the heart follows the following path-way
sinoatrial (SA) node..... atrioventricular (AV) node..... bundle of his...
bundle branches...Purkinje fiber
- Normal rhythm is characterized by the following
 - ✓ Originates from SA-node
 - ✓ Heart rate should be 60 -100beats/m
 - ✓ Should follow the normal conduction path way
 - ✓ Should propagate with normal velocity
- ❖ *Arrhythmias represent electrical activity that deviates from the above description as a result of an abnormality in impulse initiation and/or impulse propagation*

Classification of arrhythmia

- ❖ Based on heart beat abnormality
- ✓ **Tachyarrhythmia (Tachycardia)**

When $hb > 100 \text{ beats/min}$

Can be classified as

Simple, paroxysmal, flutter and fibrillation

- ✓ Depending on the site of origin, the tachyarrhythmias can be classified into: Atrial tachyarrhythmias and ventricular tachyarrhythmias
- ✓ Brady arrhythmia (bradycardia)

When $hb < 60 \text{ beats/min}$

Can be classified as mild, moderate and Sevier

Classification of arrhythmia...

- ❖ Depending the origin of the site where aryhtimia is happened it can be classified as
- ✓ Sinus, atrial, Junctional and ventricular
- ❖ Based on the position (above and below ventricle)

Supra ventricular tachyarrhythmia (sinus, atrial and junctional) vs ventricular.

BASIC PHARMACOLOGY OF THE ANTIARRHYTHMIC AGENTS

The aim of therapy of arrhythmias is

- ✓ To reduce ectopic pacemaker activity and
- ✓ Modify conduction or refractoriness
- ✓ The major pharmacologic mechanisms currently available for accomplishing these goals are
 - (1) sodium channel blockade,
 - (2) blockade of sympathetic autonomic effects in the heart,
 - (3) prolongation of the effective refractory period, and
 - (4) calcium channel blockade.

- ❖ The most widely used scheme for the classification of antiarrhythmic drug actions recognizes four classes:
 - ❖ Class 1 action is sodium channel blockade (Class 1A, 1B and 1C).
 - ❖ Class 2 action is sympatholytic
 - ❖ Class 3 action manifests as prolongation of APD
 - ❖ Class 4 action is blockade of the cardiac calcium current
- ❖ Failure of this classification system
- ✓ A given drug may have multiple classes of action
e.g amiodarone
- ✓ . Certain antiarrhythmic agents do not fit in to any of these classes
E.g. Adenosine and Magnesium

Class 1 action is sodium channel blockade

Class 1A

Procainamide

Cardiac effect

- ✓ slows the upstroke of the action potential,
- ✓ slows conduction,
- ✓ Prolongs the QRS duration of the ECG.
- ✓ The drug also prolongs the APD (a class 3 action) by nonspecific blockade of potassium channels.
- ❖ The drug may be somewhat less effective than quinidine in **suppressing abnormal ectopic pacemaker** activity but more effective in blocking sodium channels in depolarized cells.

Procainamide...

Extracardiac Effects

- ❖ hypotension, particularly with intravenous use. excessively rapid procainamide
- ✓ hypotension is associated with rapid infusion or the presence of severe underlying left ventricular dysfunction.

Toxicity

- ❖ Excessive action potential prolongation,
- ❖ Qt-interval prolongation, and
- ❖ Induction of torsades de pointes arrhythmia and syncope.
- ❖ Excessive slowing of conduction can also occur.

Pharmacokinetics & Dosage

- ❖ Procainamide can be administered
- ✓ By **intravenous** and **intramuscular** routes and is well absorbed **orally**.

A metabolite (*N*-acetylprocainamide, NAPA) has class 3 activity.

- ✓ has been implicated in torsades de pointes during procainamide therapy, especially in patients with renal failure.
- ✓ Procainamide is eliminated by hepatic metabolism to NAPA and by renal elimination.
- ✓ Its half-life is only 3–4 hours, which necessitates frequent dosing or use of a slow-release formulation (the usual practice).
- ✓ NAPA is eliminated by the kidneys.

Procainamide

- If a rapid procainamide effect is needed, an intravenous loading dose of up to 12 mg/kg can be given at a rate of 0.3 mg/kg/min or less rapidly.
- ❖ Procainamide is effective against most **atrial** and **ventricular** arrhythmias

Quinidine

❖ **Cardiac Effects**

- ✓ Quinidine has actions similar to those of procainamide
- ❖ Toxic cardiac effects include
 - ✓ Excessive QT-interval prolongation
 - ✓ induction of torsades de pointes arrhythmia.
 - ✓ It also has modest **antimuscarinic** actions in the heart.

Extracardiac Effects

- ✓ Diarrhea,
- ✓ Nausea
- ✓ Vomiting
- ✓ **Cinchonism**

Disopyramide

Cardiac Effects

- ✓ The effects of disopyramide are very similar to those of procainamide and quinidine.
- ✓ antimuscarinic effects >quinidine.
 - a drug that slows AV conduction should be administered with disopyramide when treating **atrial flutter** or **fibrillation**.

Toxicity

- ✓ Toxic concentrations results in disturbances described under quinidine. As a result of its negative inotropic effect, disopyramide may precipitate heart problem
- ✓ It should not be used in patients with heart failure.
- ✓ **Disopyramide's atropine-like** activity accounts for most of its symptomatic adverse effects:
- ✓ .These effects may require discontinuation of the drug.

Disopyramide

Pharmacokinetics & Dosage

- ✓ Oral dosage of disopyramide is 150 mg three times a day, -1 g/d has been used.

Dosage should be reduced

- ✓ In patients with renal impairment,
- ✓ .Because of the danger of **precipitating heart failure**, loading doses are **not recommended**.

Therapeutic Use

It is approved only for the treatment of **ventricular arrhythmias**

Group 1B

Lidocaine

- ✓ Has a low incidence of toxicity
- ✓ high degree of effectiveness in arrhythmias associated with acute **myocardial infarction**.
- ✓ It is used only by the intravenous route.

Cardiac Effects

- blocks activated and inactivated sodium channels with rapid kinetic
- the inactivated state block ensures greater effects on cells with long action potentials such as **Purkinje** and **ventricular cells**, compared with **atrial cells**.
- In depolarized cells, the increased inactivation results in **selective depression of conduction**.
- **Little effect is seen** on the **ECG** in normal sinus rhythm.

Lidocaine...

Toxicity

- Lidocaine is one of the **least cardiotoxic** of the currently used sodium channel blockers.
- Proarrhythmic effects(**SA node arrest, worsening of impaired conduction**, and ventricular arrhythmias, are uncommon with lidocaine)
- Larger dose.....hypotension
most common adverse
- neurologic: paresthesias, tremor, nausea of central origin, lightheadedness, hearing disturbances, slurred speech, and convulsions.
In elderly or otherwise vulnerable patients or when a bolus of the drug is given too rapidly
- ✓ seizures respond to intravenous diazepam. If plasma level <9 mcg/mL
.....lidocaine is well tolerated.

Lidocaine...

Pharmacokinetics & Dosage

- ✓ **only 3%** of orally administered lidocaine appears in the plasma....First pass effect.
- ✓ lidocaine must be **given parenterally**.
- ✓ half-life of 1–2 hours.

Dose

- ✓ loading dose of 150–200 mg administered over about 15 minutes (as a single infusion or as a series of slow boluses) followed by a maintenance infusion of 2–4 mg/min to achieve a therapeutic plasma level of 2–6 mcg/mL.
- ✓ In patients with heart failure, lidocaine's volume of distribution and total body clearance may both be decreased.

Lidocaine...

For management of

- ✓ **ventricular tachycardia** and prevention of **ventricular fibrillation** after cardioversion in the setting of acute ischemia.

Mexiletine

- ✓ is an orally active congener of lidocaine.
- ✓ **antiarrhythmic actions** are similar to those of **lidocaine**
 - The anticonvulsant phenytoin exerts similar electrophysiologic effects and has been used as an antiarrhythmic.)

Pharmacokinetics & Dosage

- ✓ The elimination half-life is **8–20**....administered two or three times. The usual daily dosage of mexiletine is 600–1200 mg/d.

Mexiletine

Toxicity

- ✓ neurologic, including tremor, blurred vision, and lethargy and Nausea is also a common effect.
- ✓ Non- cardiovascular uses
- ✓ In relieving chronic pain, especially pain due to diabetic neuropathy and nerve injury.

Dosage is 450–750 mg/d orally. This application is off label.

- ✓ Mexiletine is used in the treatment of **ventricular arrhythmias**

Group 1C

Flecainide

It is a potent blocker of **sodium** and **potassium** channels with slow unblocking kinetics.

- ✓ It is currently used for patients with **supraventricular arrhythmias**.
- ✓ **suppressing premature ventricular** contraction but exacerbate **ventricular tachyarrhythmias**
- ✓ It has no **antimuscarinic** effects.

Dosage and administration

- ✓ Flecainide is well absorbed
- ✓ half-life ----20 hours.
- ✓ Elimination is both by hepatic metabolism and by the kidney.
- ✓ The usual dosage of flecainide is 100–200 mg twice a day.

Moricizine

- ✓ It is an antiarrhythmic phenothiazine derivative that was used for treatment of **ventricular arrhythmias**.
- ✓ It is a relatively potent **sodium channel blocker** that **does not prolong action potential duration**.
- ✓ Moricizine has been withdrawn from the US market.

Beta-adrenoceptor-blocking drugs (class 2)

- ✓ They are well tolerated,
- ✓ Their efficacy for suppression of **ventricular ectopic depolarizations** is lower than that of sodium channel blockers.

Pharmacological uses

- ⊙ Treatment of increased sympathetic activity-induced arrhythmias such as stress- and exercise-induced arrhythmias
- ⊙ Atrial flutter and fibrillation.
- ⊙ AV nodal tachycardia.
- ⊙ Reduce mortality in **post-myocardial infarction** patients
- ⊙ Protection against sudden cardiac death

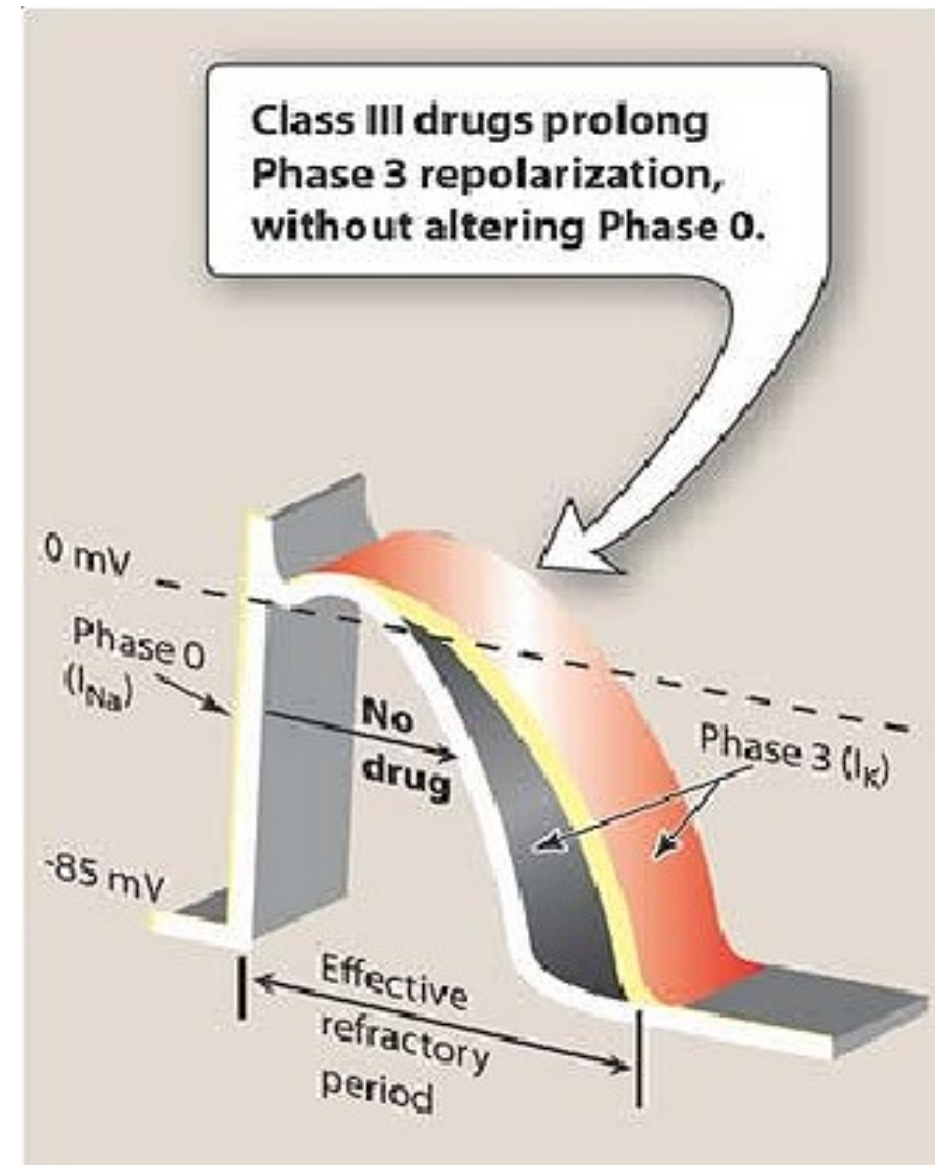
Beta-adrenoceptor-blocking drugs (class 2)...

- **Propranolol (nonselective):** was proved to reduce the incidence of sudden arrhythmic death after myocardial infarction
- **Metoprolol**
 - reduce the risk of bronchospasm
- ***Esmolol:***
 - Esmolol is a very short-acting β_1 -adrenergic blocker that is used by intravenous route in **acute arrhythmias** occurring during surgery or emergencies
 - **Sotalol** is a nonselective β -blocking drug that prolongs the action potential (class 3 action).

Class III ANTIARRHYTHMIC DRUGS

K⁺ blockers

- Prolongation of phase 3 repolarization without altering phase 0 upstroke or the resting membrane potential
- They prolong both the **duration of the action potential** and **ERP**
- Their mechanism of action is still not clear but it is thought that they **block potassium channels**



Uses:

- Ventricular arrhythmias, especially ventricular fibrillation or tachycardia
- Supra-ventricular tachycardia
- Amiodarone usage is limited due to its wide range of side effects

Sotalol (Sotacor)

- ✓ Sotalol also prolongs the duration of action potential and refractoriness in all cardiac tissues (by action of K^+ blockade)
- ✓ Sotalol suppresses Phase 4 spontaneous depolarization and possibly **producing** severe **sinus bradycardia** (by β blockade action)
- ✓ The β -adrenergic blockade combined with prolonged action potential duration may be of special efficacy in prevention of **sustained ventricular tachycardia**
- ✓ It may induce the polymorphic **torsades de pointes ventricular tachycardia** (because it increases ERP)

Ibutilide

- ❖ Used in atrial fibrillation or flutter
- ❖ IV administration
- ❖ May lead to **torsade de pointes**
- ❖ Only drug in class three that possess pure K^+ blockade

Amiodarone

- ✓ Amiodarone is a drug of multiple actions and is still not well understood
- ✓ It is extensively taken up by tissues, especially fatty tissues (extensive distribution)
- ✓ $t_{1/2} = 60$ days
- ✓ Potent P450 inhibitor
- ✓ *Amiodarone antiarrhythmic effect* is complex comprising **class I, II, III, and IV actions**
- ✓ Dominant effect: **Prolongation of action potential duration and refractoriness**
- ✓ It slows **cardiac conduction**, works as Ca^{2+} channel blocker, and as a weak β -adrenergic blocker

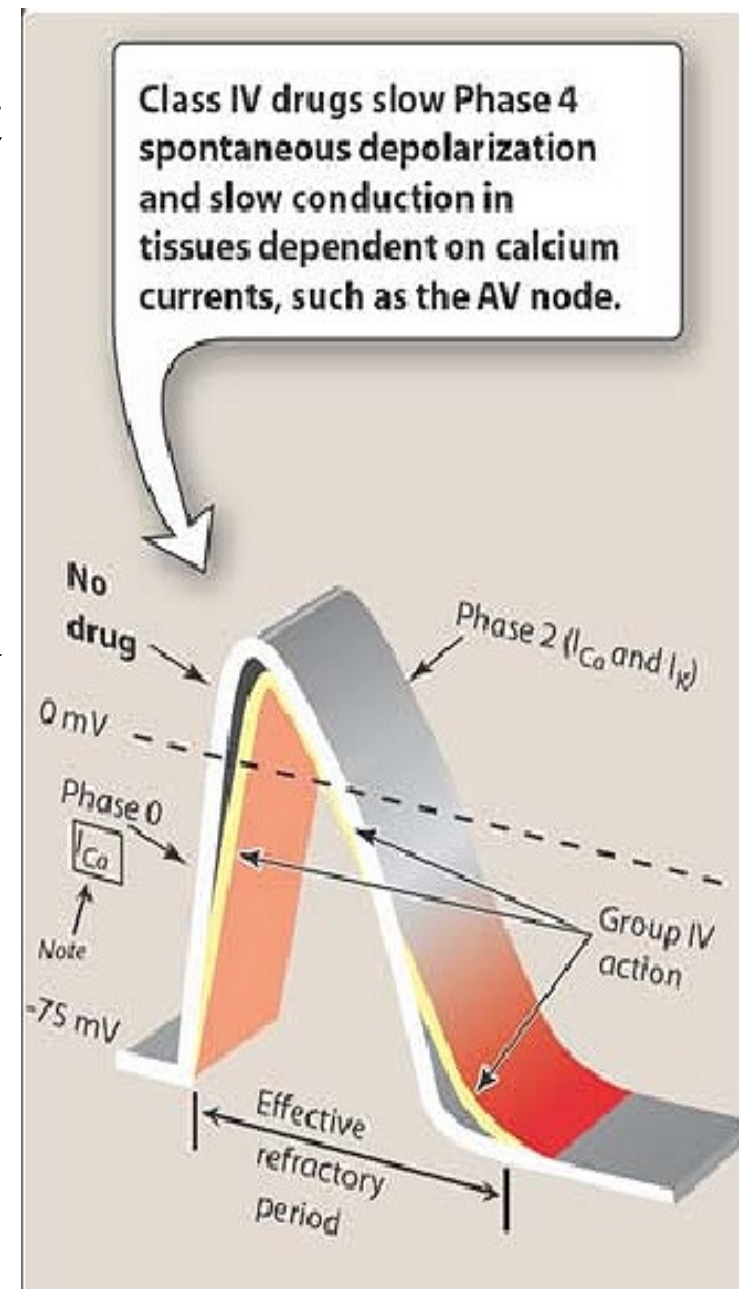
Amiodarone...

Toxicity

- Most common include GI intolerance, tremors, ataxia, dizziness, and hyper-or hypothyroidism
- **Corneal microdeposits** may be accompanied with disturbed night vision
- Others: liver toxicity, photosensitivity, gray facial discoloration, neuropathy, muscle weakness, and weight loss
- The most dangerous side effect is *pulmonary fibrosis* which occurs in **2-5%** of the patients

class IV antiarrhythmic drugs (calcium channel blockers)

- ❖ Calcium channel blockers decrease inward Ca^{2+} currents resulting in a decrease of phase 4 spontaneous depolarization (SA node)
 - ❖ They slow conductance in Ca^{2+} current-dependent tissues like AV node.
 - ❖ Examples: **verapamil & diltiazem**
- Because they act on the heart only and not on blood vessels.
- ❖ Dihydropyridine family are not used because they only act on blood vessels



Mechanism of action

- ⦿ They bind only to depolarized (open) channels prevention of repolarization
/
- ⦿ They prolong ERP of AV node ↓conduction of impulses from the atria to the ventricles

Uses

- ❑ More effective in treatment of **atrial** than **ventricular arrhythmias**.
- ❑ Treatment of supra-ventricular tachycardia preventing the occurrence of ventricular arrhythmias
- ❑ Treatment of atrial flutter and fibrillation

Contraindication

- ✓ Contraindicated in patients with pre-existing depressed heart function because of their negative **inotropic activity**

Adverse effects

- ✓ Cause bradycardia, and asystole especially when given in combination with **β -adrenergic blockers**

Miscellaneous Antiarrhythmic Drugs

Adenosine

- Adenosine **activates** A_1 -purinergic receptors **decreasing the SA nodal firing** and **automaticity**, **reducing conduction velocity**, **prolonging effective refractory period**, and **depressing AV nodal conductivity**
- It is the drug of choice in the treatment of **paroxysmal supra-ventricular tachycardia**
- It is used only by slow intravenous bolus
- It only has a low-profile toxicity (lead to bronchospasm) being extremely short acting for 15 seconds only